

Contradiction and Continuation

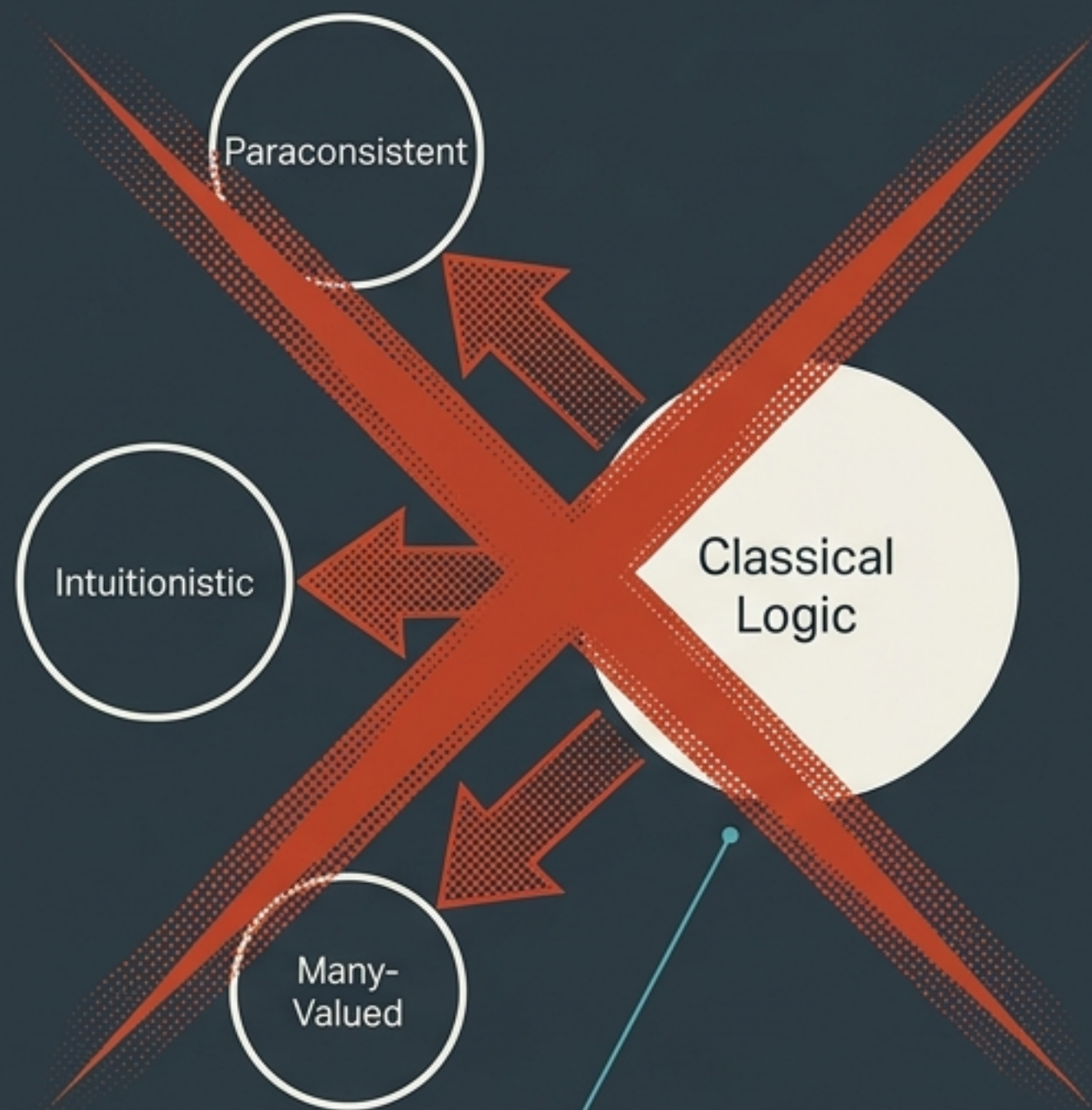
A Constraint-Theoretic Derivation
of Non-Classical Logic

Flyxion Independent Researcher

2026

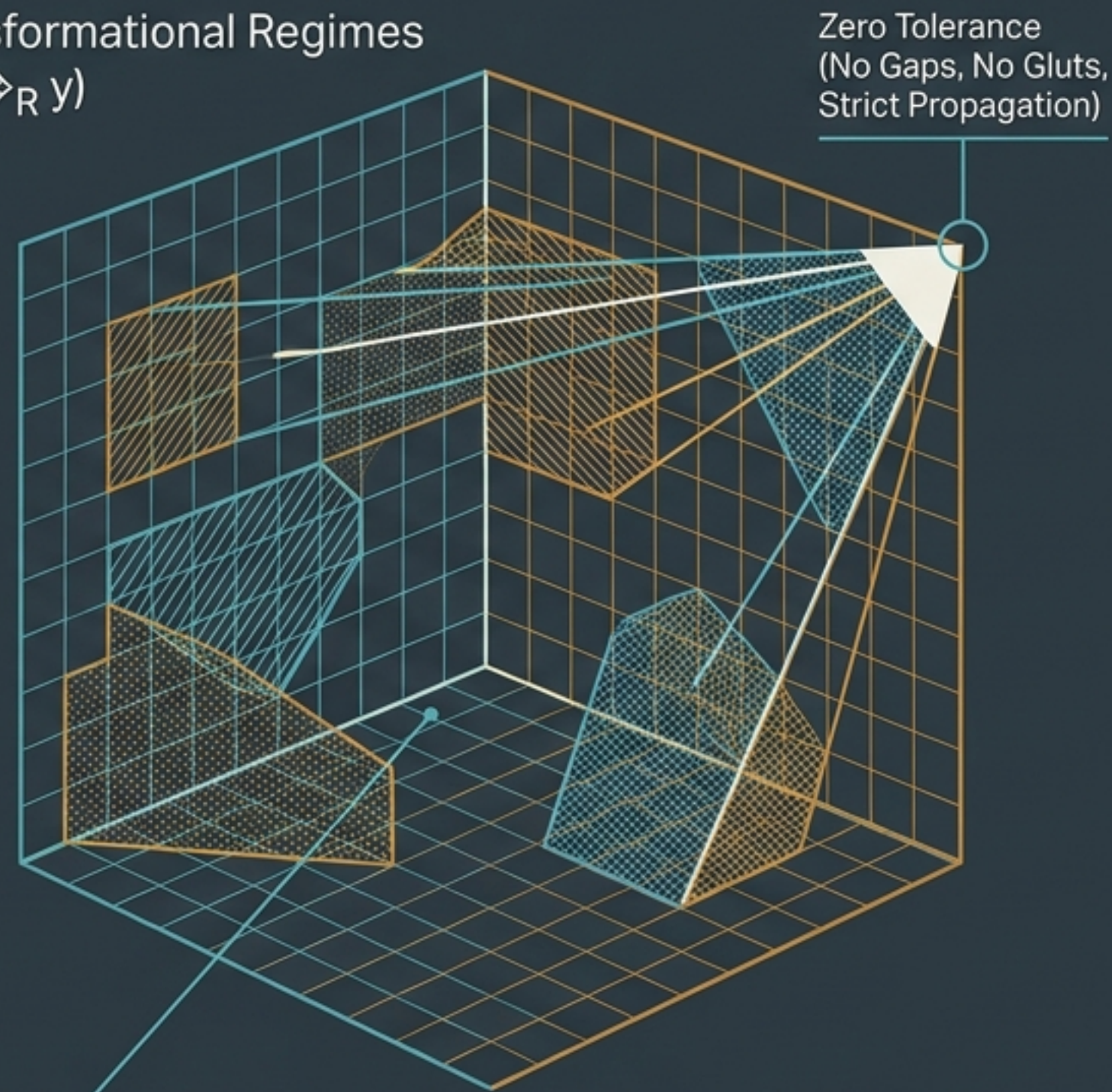
Logical consequence is preservation under
admissible transformation. Classical logic
is not the center—it is merely one point.

The Myth of the Classical Default



The old view: Classical logic is the ground state; alternatives require justification.

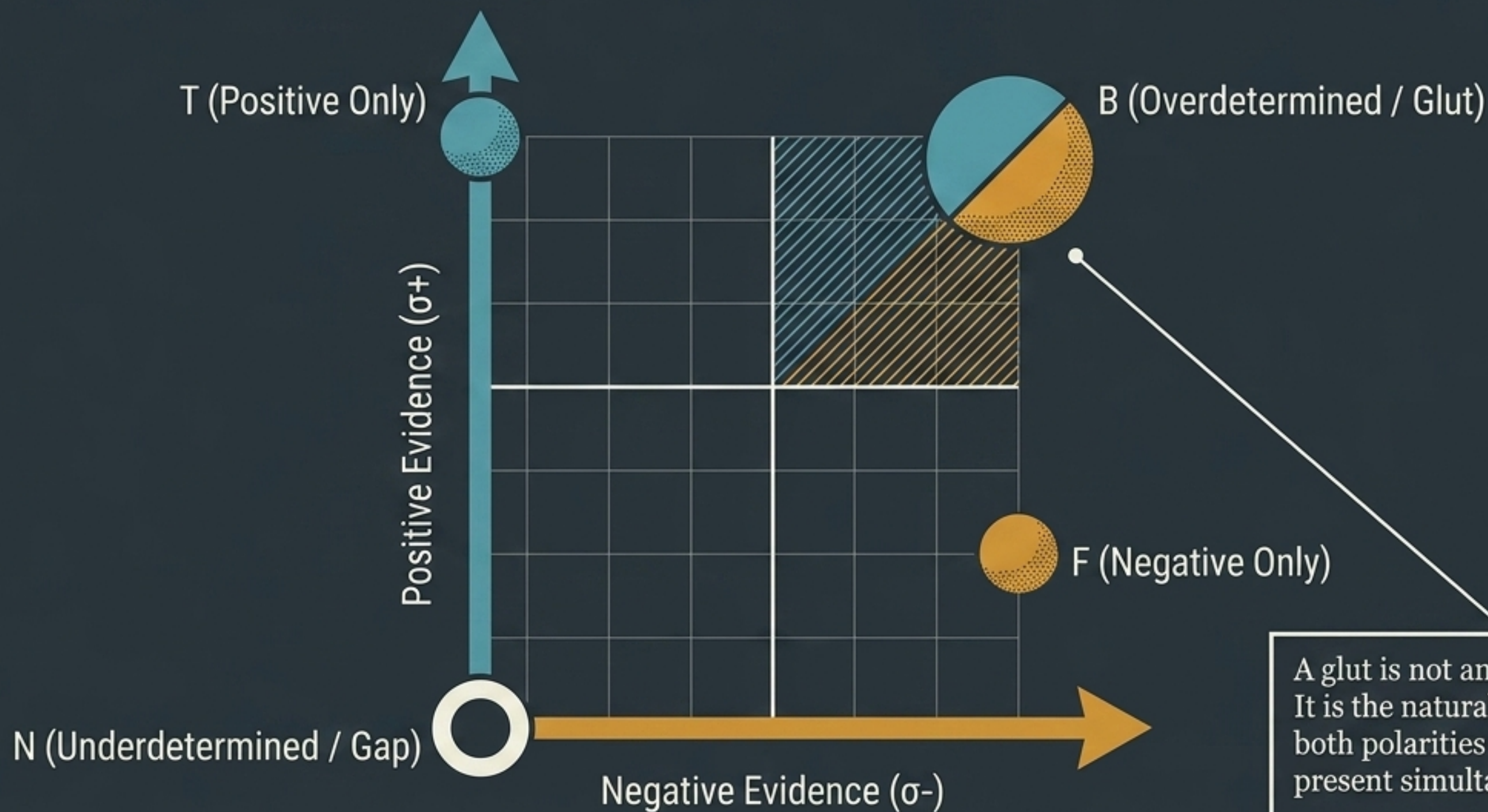
Transformational Regimes
($x \rightsquigarrow_R y$)



The geometric view: Every logic is a specific regime (R) governing what survives when representations are transported or combined. Classical logic is simply the point of maximal strictness on all axes at once.

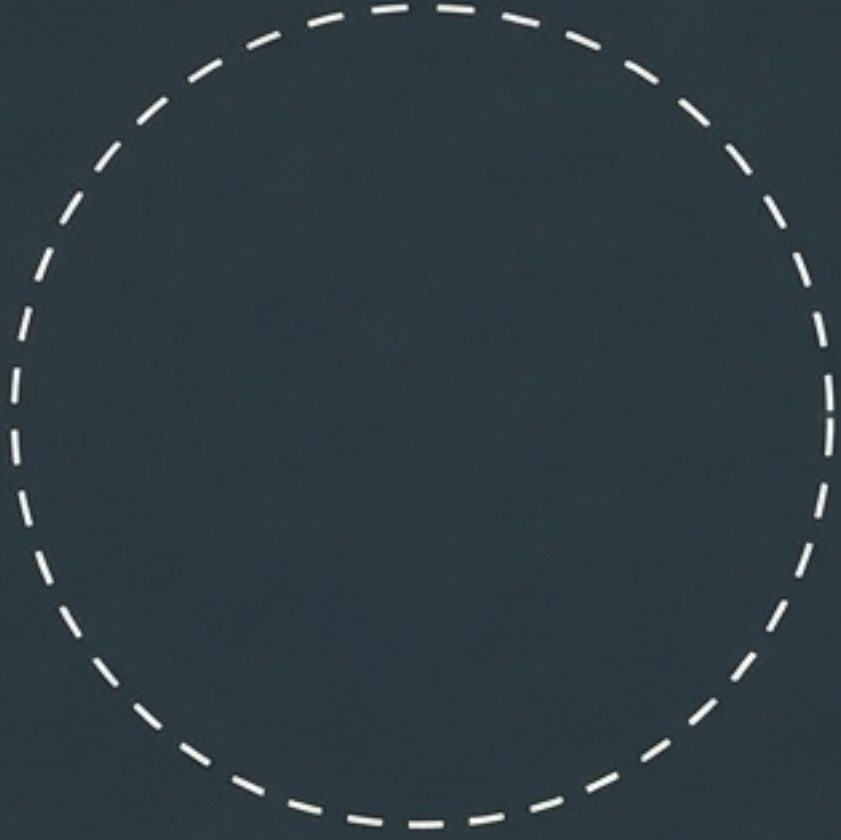
Distinction Before Truth: The Evidential Matrix

Truth values are not stipulated alphabets.
They are read off the independent
presence or absence of evidence.



A glut is not an exotic 5th value.
It is the natural consequence of
both polarities of evidence being
present simultaneously.

The Asymmetry of Gaps and Gluts



$$N \vee x = x$$

GAP (N)

Insufficient polar evidence. Harmless. Costs nothing to extend, obstructs nothing. You do not repair a gap; you fill it in.

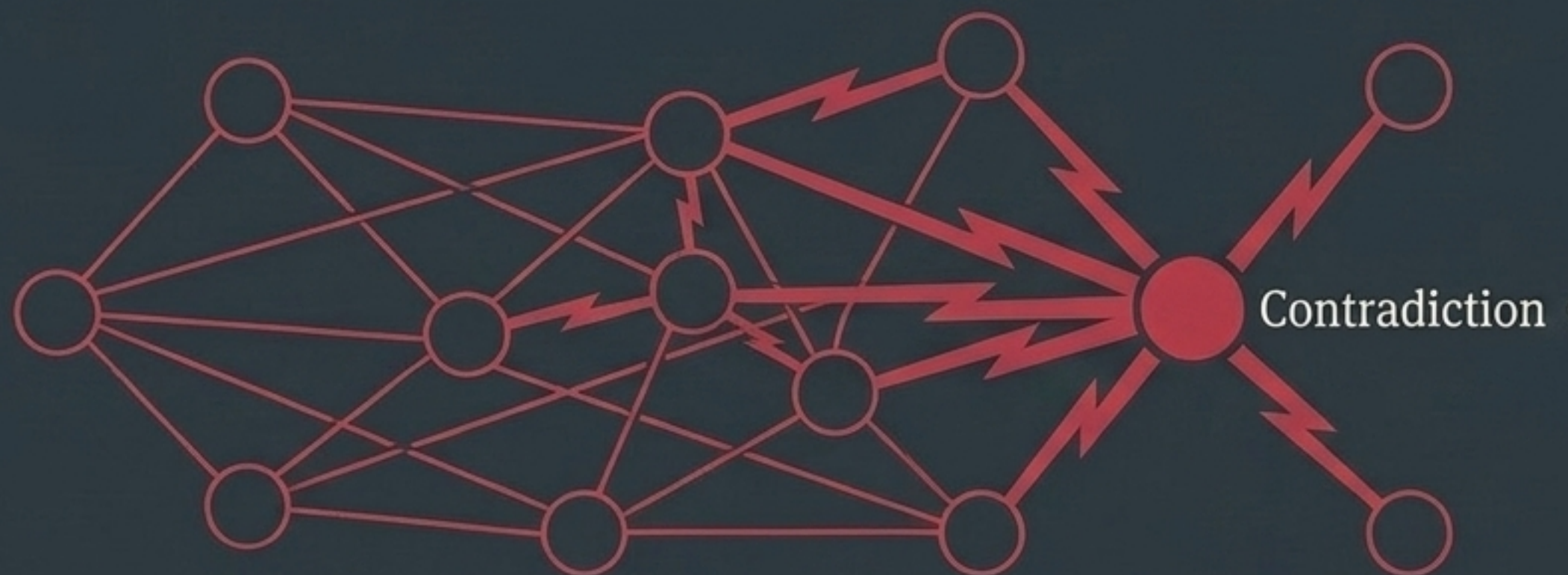


$$B \vee x = B$$

GLUT (B)

Incompatible polar evidence. The substantive case. A structural conflict that requires a principled repair strategy and restricted propagation.

Paraconsistency: Containing the Blast Radius



Ex Falso Quodlibet

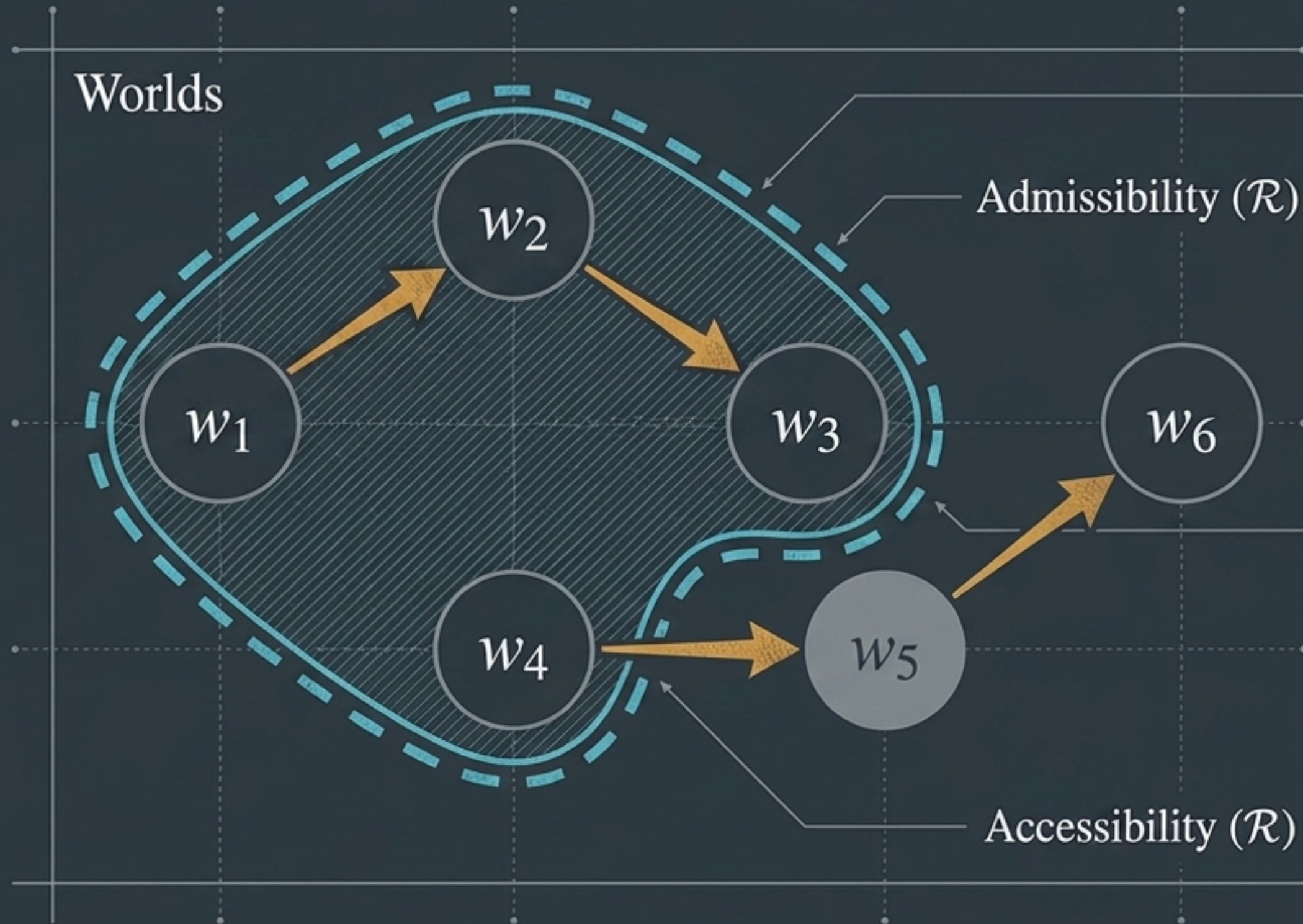
Classical logic routes local failure into global death. Inconsistency
Inconsistency = Triviality.



Restricted Closure

Paraconsistent regimes withhold the specific closure rule that connects local conflict to global collapse. Contradiction is treated as retained information, not systemic failure.

Splitting the Modal Atom



The Conflation

Traditional modal logic conflates “which states satisfy the rules” with “which states a given world can currently reach.”

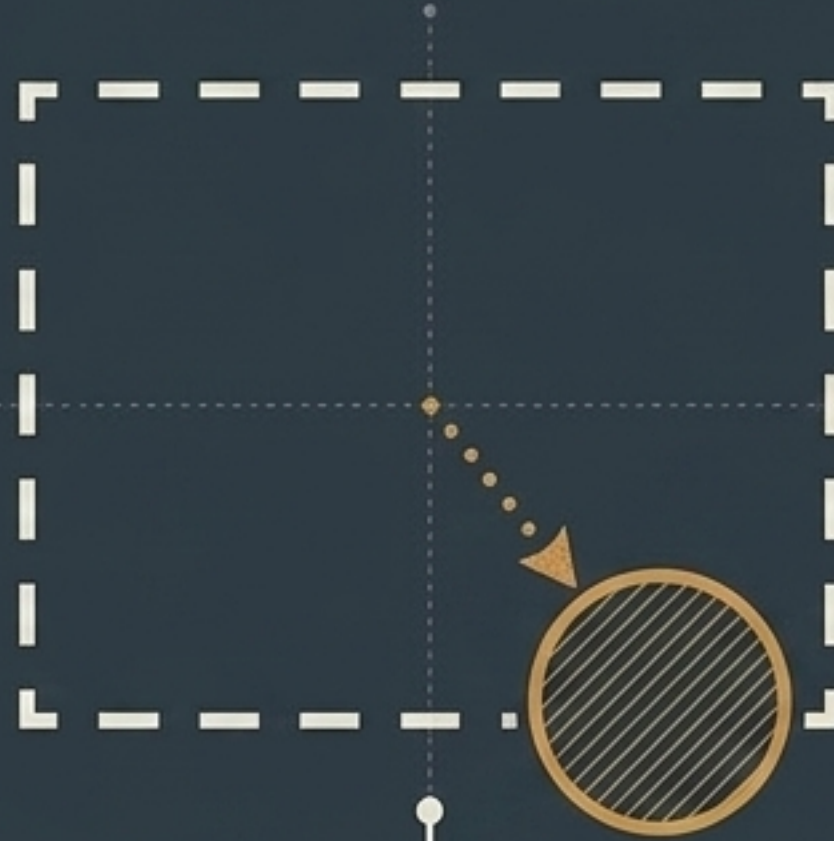
The Separation

We must split $\mathbb{R}$ and $\mathcal{R}$.
Admissibility (the laws of the regime) does not dictate
Accessibility (the vantage point).
A world can be accessible but inadmissible.

Suspension, Not Refusal



Inconsistency (Support)
Contradiction is a semantic fact indexed to support.



Impossibility (Constraint)
Impossibility is a structural fact indexed to a constraint regime.



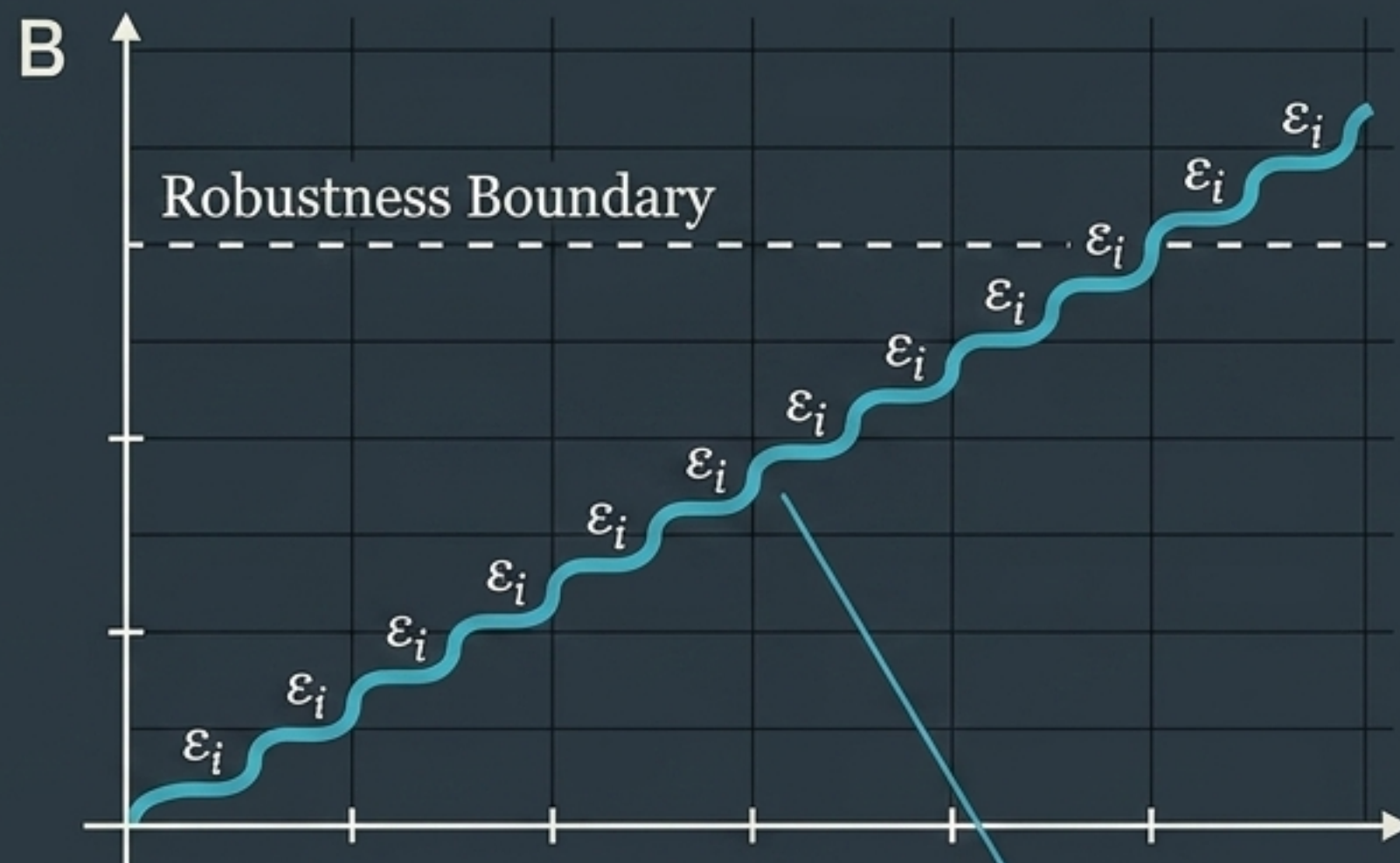
Non-Normality (Suspension)
Non-normal worlds don't just refuse a single rule—the evaluation rule itself is suspended (a state-indexed regime, Rw).

Vagueness: Dispersion vs. Obstruction



Compatibility Obstruction

Local disagreement. A structural break between adjacent data blocks.

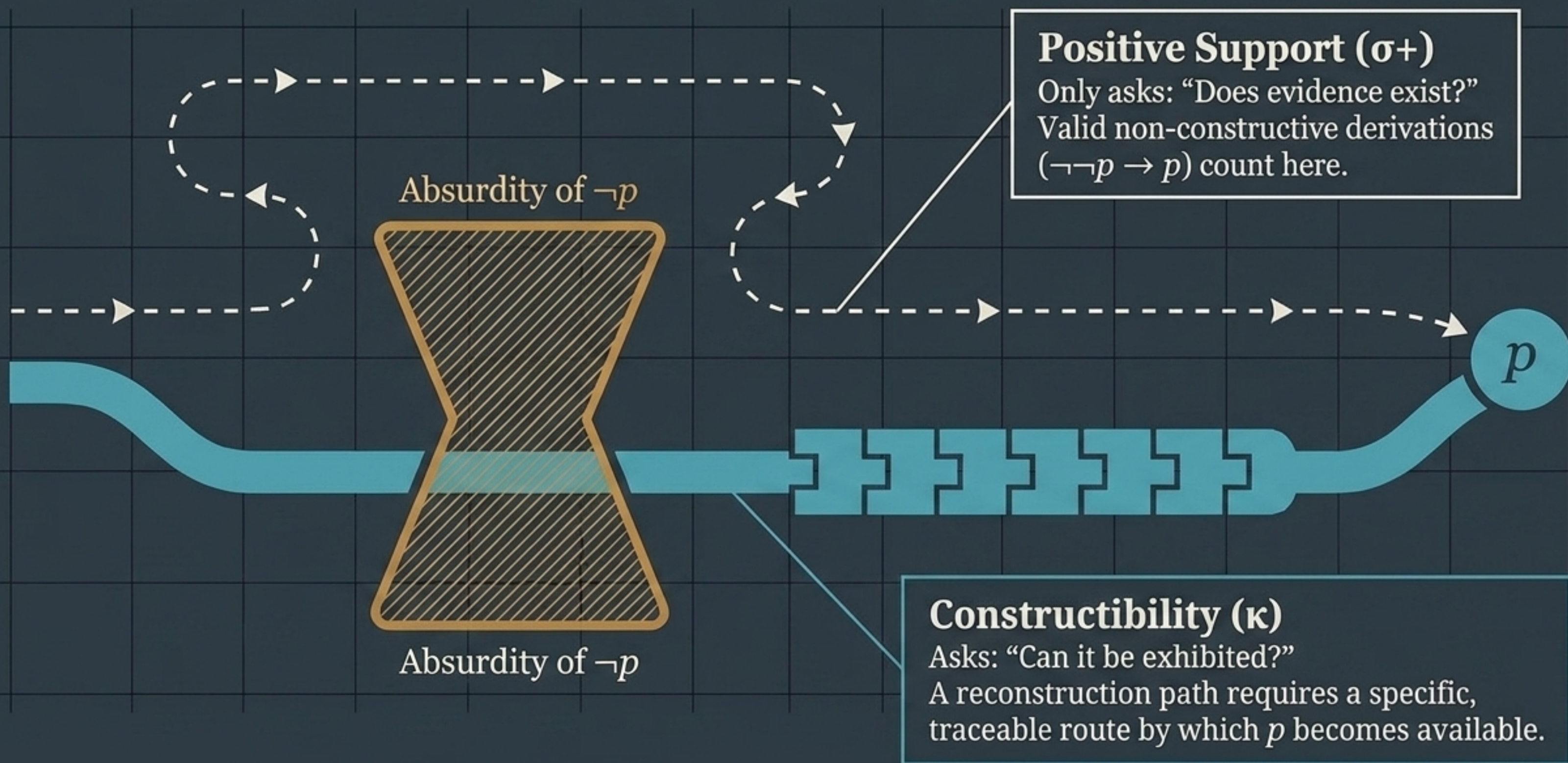


Dispersion (ρ)

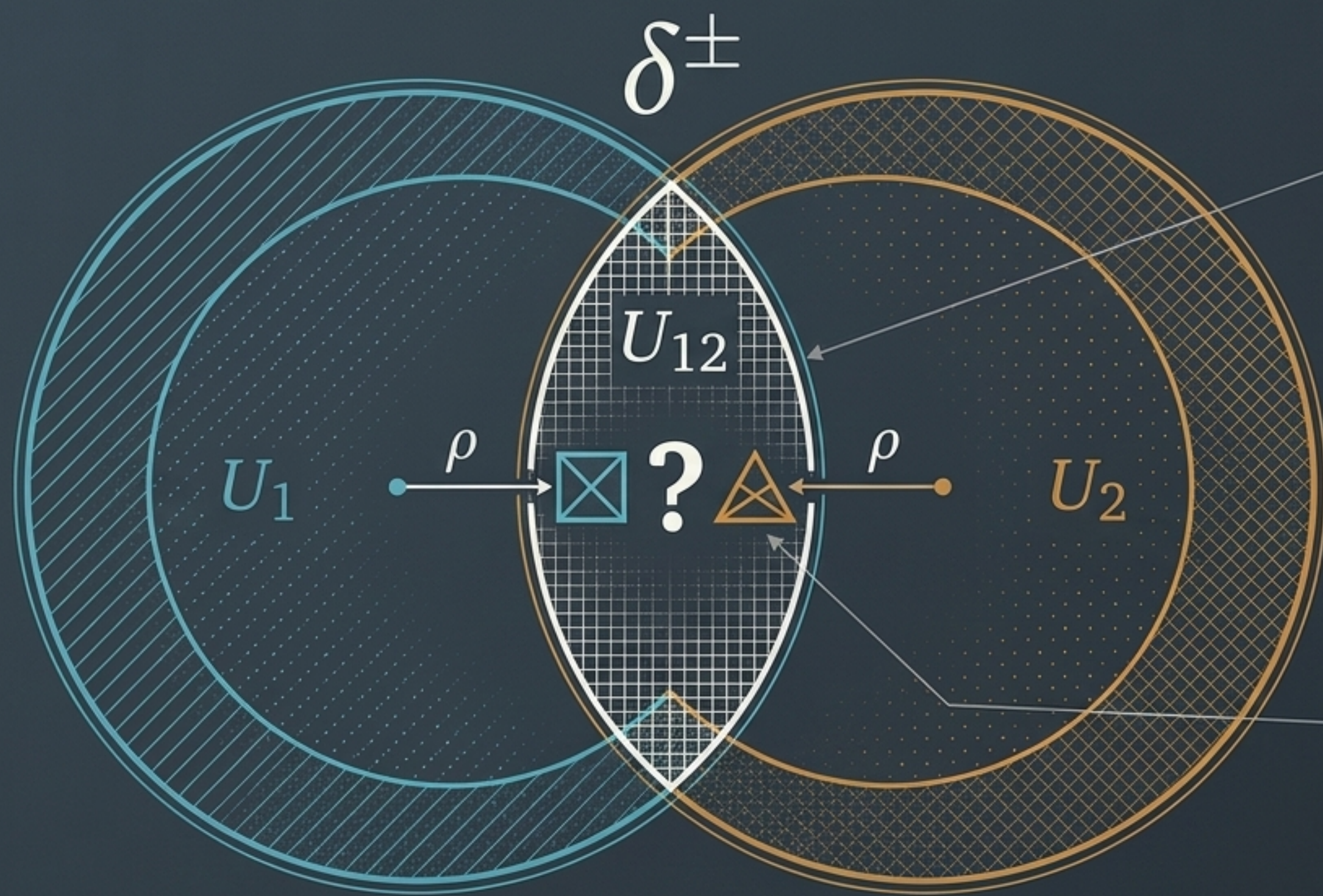
Local smallness fails to remain globally small under iteration ($\sum \epsilon_i \geq L$).

Compatibility $\neq$ Continuity $\neq$ Persistence.

Truth as Construction



The Local-to-Global Leap: Coherence Defect



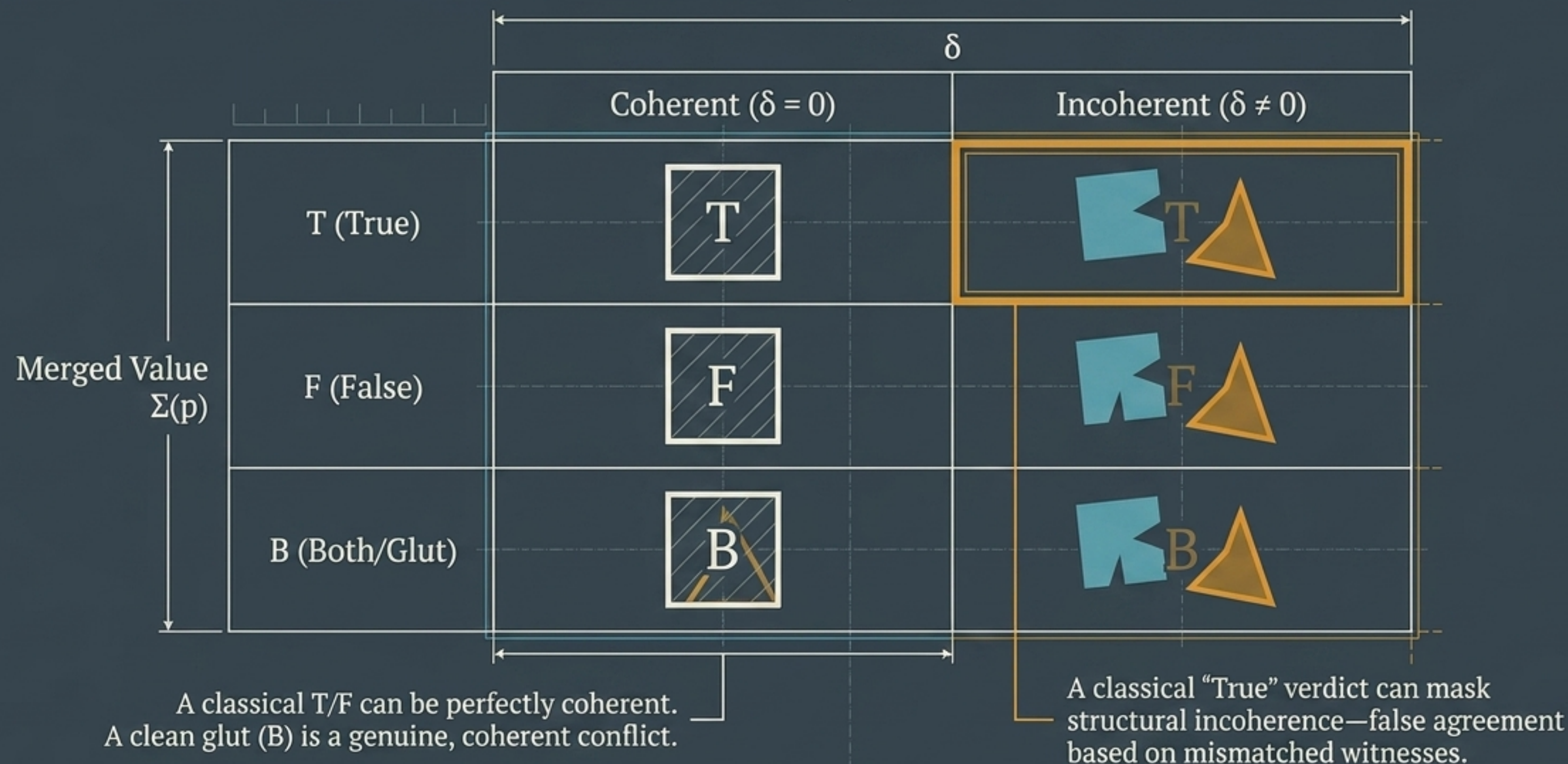
Core Question:

If Patch 1 and Patch 2 both agree a proposition is "True," do they agree on the same underlying witness?

Insight:

The Coherence Defect ($\delta_\pm$) measures if overlapping patches contribute the exact same evidence. Agreement on a verdict is not agreement on reality.

Contradiction $\neq$ Obstruction



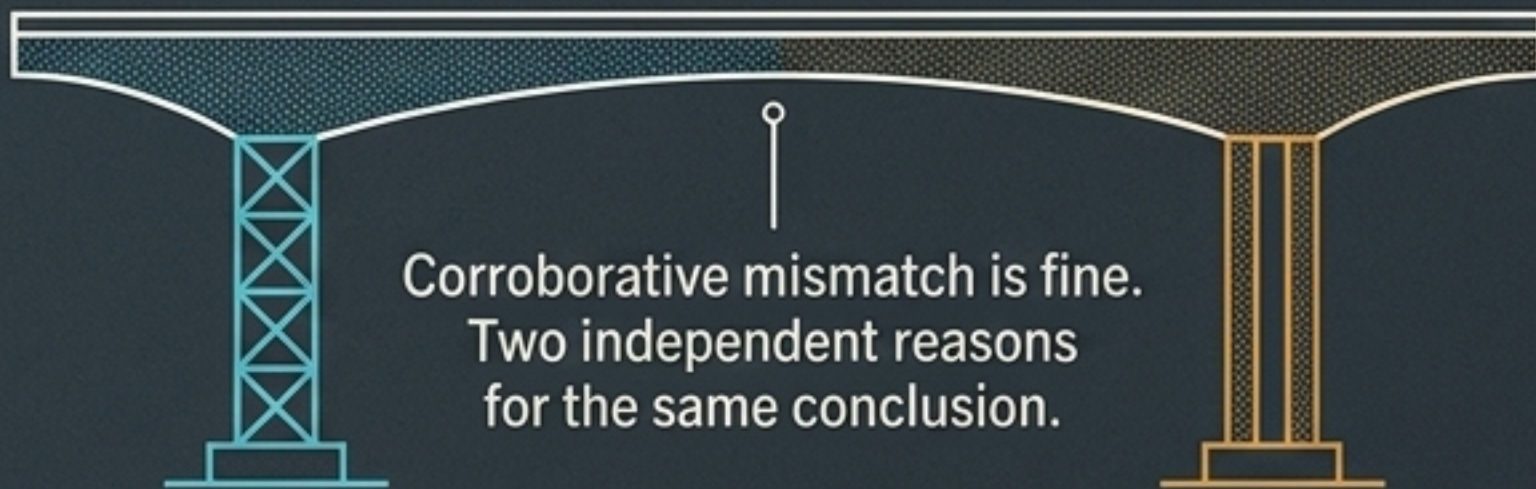
Contradiction is a semantic fact about support. Obstruction is a geometric fact about coherence.
Four-valued logic alone is blind to this.

The Third Term: Action & Liability

$$J(a) = \left[\begin{array}{l} \text{(Action Cost)} \\ E(a) + \\ \lambda R(a) \end{array} \right] \begin{array}{l} \text{(Base effort)} \\ \text{(Repairing a Glut)} \end{array} \quad \downarrow \text{Old model ends here.}$$

$$+ \left[\mu C(a) \right] \quad \text{Coherence Liability}$$

Bridge

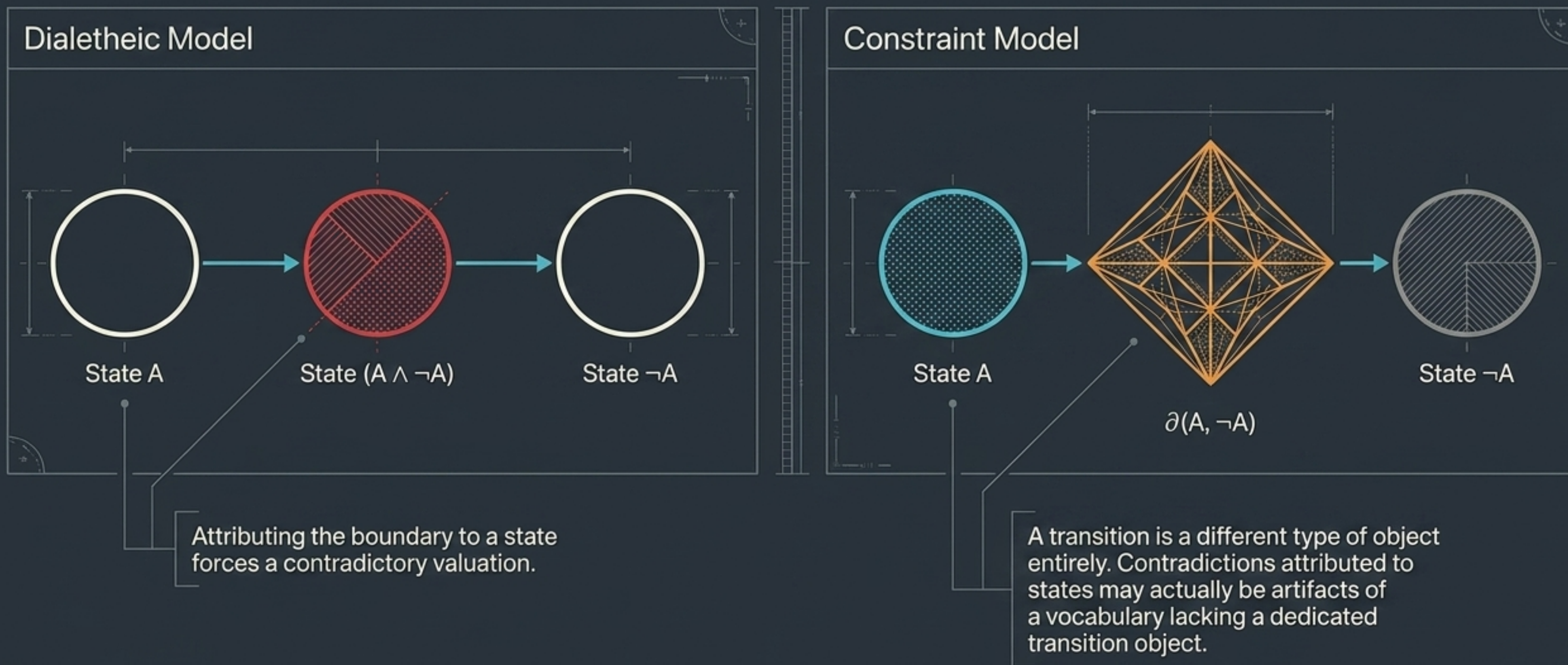


Keycard

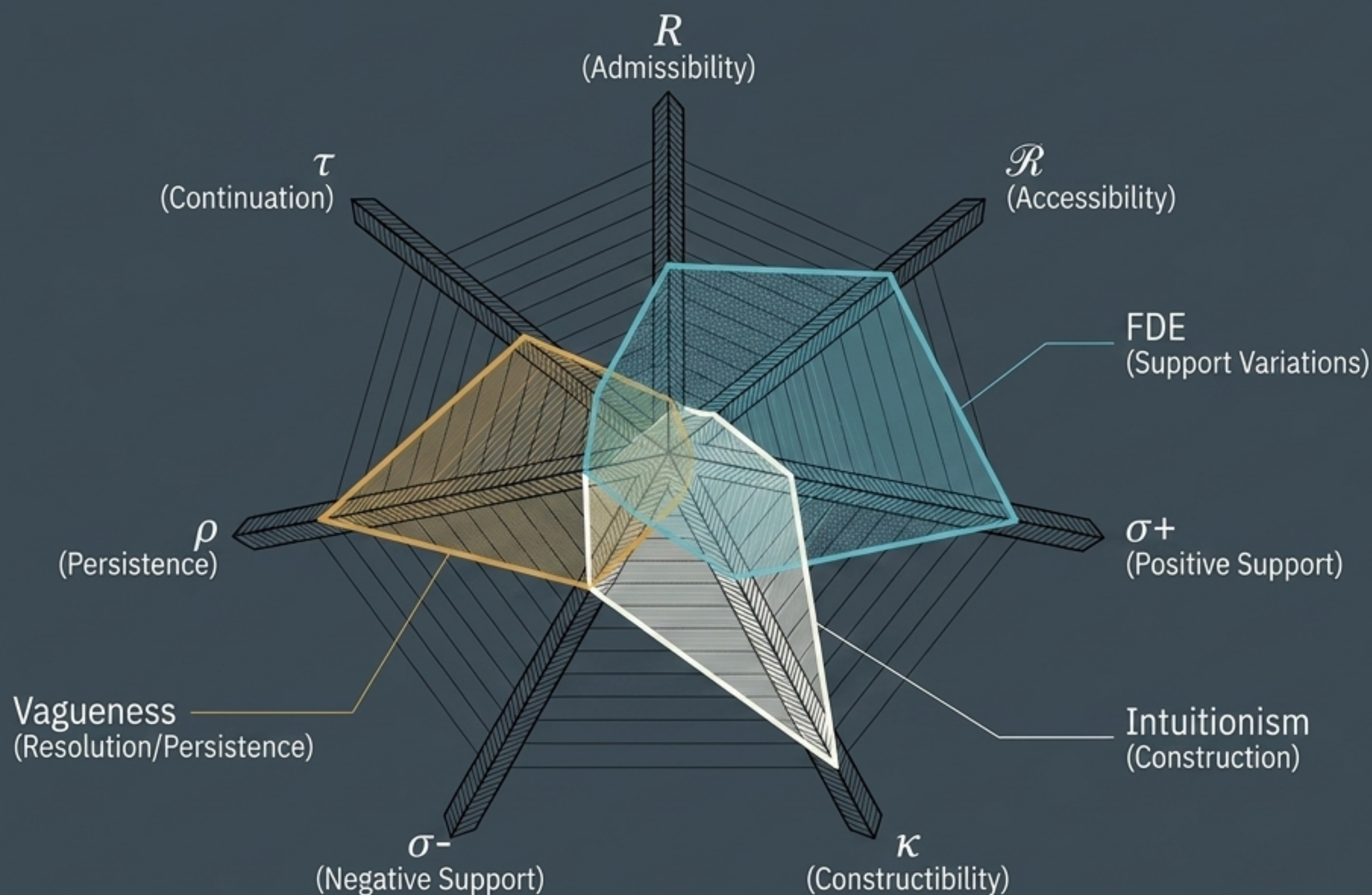


Rational action requires a third term. Cost depends entirely on what the specific action needs from the evidence.
Logical agreement does not imply evidential coherence.

Identity Under Transport & The Transition Object



Synthesis: The Geometry of Logical Regimes

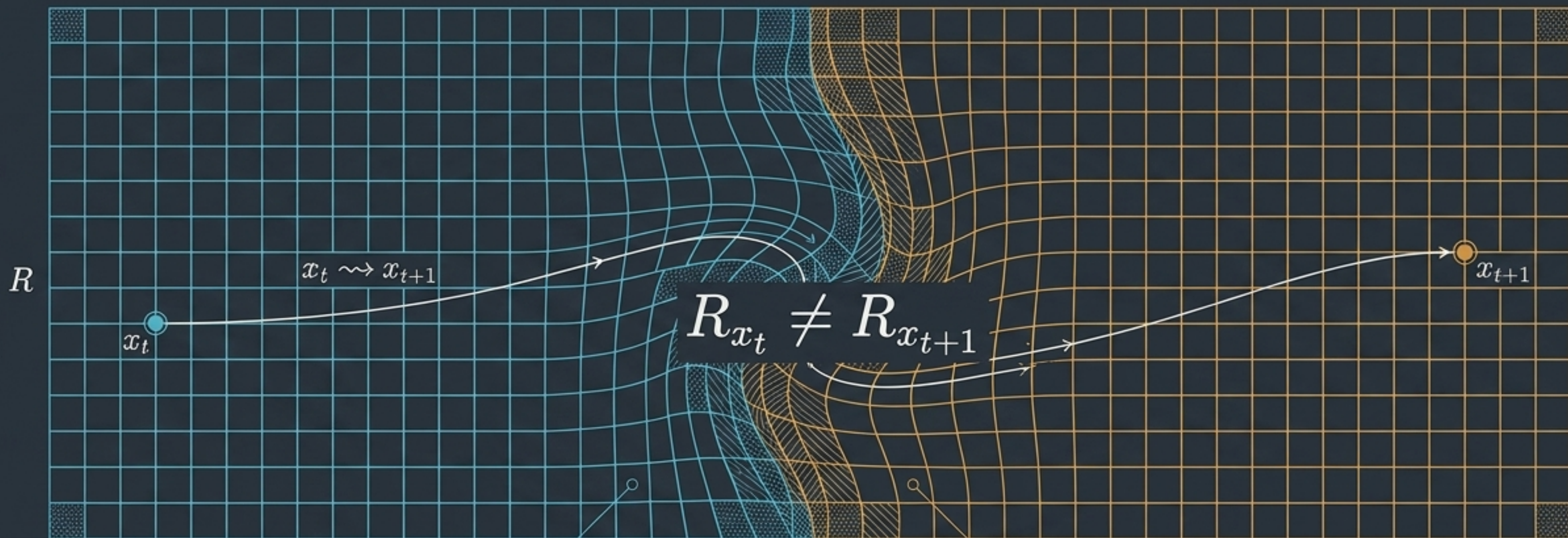


Core Concept:

The taxonomy of logic becomes a geometry.

Non-classical logics are not competing rulesets;
Non-classical logics are not competing rulesets;
they are mathematical sub-regions within a unified Support-Coherence Space.

The Frontier: Evolving Regimes



The Open Problem

What happens when the regime governing admissibility varies along the trajectory it is supposed to govern?

Closing Thought

To understand identity across profound change, we need an account of what makes a transition between regimes admissible—not just what makes endpoints admissible on their own.